

Unit 1 - Foundations of Computing and Digital Systems

1 Computing in Everyday Life

How do computers process information, and what role does logic play in their decision-making?

2 Introduction to Computing

How has computing evolved from early mechanical systems to AI-driven automation?

Case study - compare and contrast Colossus vs my laptop

“Colossus incorporated 1500 valves, and operated in parallel arithmetic mode at 5000 pulses per second. Some of its features have been disclosed, including the fact that it had prodigiously fast (5000 characters per second) paper tape inputs, electronic counting circuits, binary arithmetic and Boolean logic operations, “electronic storage resistors changeable by an automatically controlled sequence of operations”, “conditional (branching) logic”, “logic functions pre-set by patchpanels or switches, or conditionally selected by telephone relays”, and typewriter output.” (Randell, 1982)

Feature	Colossus (1943–44)	My HP OMEN Transcend 14
Era	World War II	2025
Primary purpose	Codebreaking (Lorenz cipher)	General-purpose computing
Programmability	Semi-programmable via switches & patch panels	Fully programmable (software)
Logic system	Binary, Boolean logic	Binary, Boolean logic
Arithmetic mode	Parallel arithmetic (fixed function)	Parallel, superscalar, vectorized

Clock / speed	~5,000 pulses per second	~2.5–5+ billion cycles per second
Processing units	Single electronic processing system	16 CPU cores / 22 threads
Components	~1,500 vacuum valves (tubes)	Billions of transistors
Physical size	Room-sized	Portable laptop
Power consumption	Kilowatts	Tens of watts
Memory (working storage)	Electronic counters & resistors	32 GB RAM
Program storage	None (hardware-configured)	Software on SSD
Input	Paper tape (5,000 chars/sec)	Keyboard, mouse, touchpad, network
Output	Typewriter / printer	High-resolution display, audio, files
Branching logic	Conditional (limited)	Fully dynamic branching
Reconfiguration time	Minutes to hours	Milliseconds
Reliability	Frequent component failures	High stability
Virtualisation	None	Hardware-assisted virtualisation
Security features	None	Secure Boot, VBS, HVCI

(OpenAI, 2026)

What are the key differences between classical and quantum computing?

Classic computing is 1 or 0, quantum computing is 1, 0 or both.

Quantum computing is still experimental and in the testing stages where as classic computing is established.

Quantum computing can theoretically iterate more problems and possible solutions.

How do computing technologies shape industries like healthcare, finance, and cybersecurity?

Cybersecurity wouldn't exist without computers... It wouldn't be needed.

Healthcare, data analysis is a huge part. Store manage and calculate individual and large groups of data. It can be used for meta analysis.

Additionally early screen of scans and images for cancer and other conditions.

Finance: instantaneous exchange of data, funds, stock prices, and trades. Communication across continents in real time.

Increase in data break and cyber attacks.

Invention of new forms of currency that are unregulated, such as bitcoin.

3 Logic, Gates and Boolean Algebra

Basic Logic Gates (AND, OR, NOT, XOR, NAND, NOR).

Truth Tables and Boolean Expressions.

AND $\wedge$

Both must be 1

$A \cdot B$



A B Output

0 0 0

0 1 0

1 0 0

1 1 1

OR $\vee$

At least one is 1

$A + B$



A	B	Output
---	---	--------

0	0	0
---	---	---

0	1	1
---	---	---

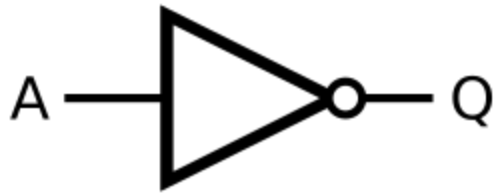
1	0	1
---	---	---

1	1	1
---	---	---

NOT $\neg$

Flips input

$\neg A$ or A'



A Output

0 1

1 0

XOR

Must be different

$A \oplus B$



A B Output

0 0 0

0 1 1

1 0 1

1 1 0

NAND

Opposite of AND - both 1 = 0 otherwise 1

$\neg(A \cdot B)$



A	B	Output
---	---	--------

0	0	1
---	---	---

0	1	1
---	---	---

1	0	1
---	---	---

1	1	0
---	---	---

NOR

Opposite of OR: both must be 0, otherwise 0

$\neg(A + B)$



A	B	Output
---	---	--------

0	0	1
---	---	---

0	1	0
---	---	---

1	0	0
---	---	---

1	1	0
---	---	---

Order, logic

Brackets, $\neg$, $\wedge$, $\vee$,

How logic gates form the basis of microprocessors.

Activity: Students will construct truth tables for simple Boolean expressions and determine outputs for given digital circuits.

Y=NAND(A,B) AND (B XOR C)					
A	B	C	NAND	B XOR C	Result
0	0	0	1	0	0
0	0	1	1	1	1
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	0	1	0
1	1	1	0	0	0

Y=NOR(A,B) XOR (A AND C)					
A	B	C	NOR A B	A AND C	Result
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	0	0	0
1	1	1	0	1	1

Y=((A XOR B) AND (C NOR D)) OR (NAND(A,C) AND ¬B)										
A	B	C	D	A XOR B	C NOR D	NAND(A,C)	NOT B	Term1	Term2	Y
0	0	0	0	0	1	1	1	0	1	1
0	0	0	1	0	0	1	1	0	1	1
0	0	1	0	0	0	1	1	0	1	1
0	0	1	1	0	0	1	1	0	1	1
0	1	0	0	1	1	1	0	1	0	1
0	1	0	1	1	0	1	0	0	0	0

0	1	1	0	1	0	1	0	0	0	0
0	1	1	1	1	0	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	0	1	1	0	1	1
1	0	1	0	1	0	0	1	0	0	0
1	0	1	1	1	0	0	1	0	0	0
1	1	0	0	0	1	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0
1	1	1	1	1	0	0	0	1	0	1

4 Set Theory in Computing (20 minutes)

What are sets, subsets, intersections, and unions in computing?

Set - collection of things: [Dog, Cat, Rabbit, Fox]

$\subseteq$ **Sub Set** - Part of another collection: [Cat, Fox]

$\cap$ **Intersections** - things in both: [Cat, Dog, Fox] [Dog, Rabbit, Cat] = [Dog, Cat]

$\cup$ **Unions** - two sets joined without duplication: [Cat, Dog, Fox] [Dog, Rabbit, Cat] = [Dog, Cat, Rabbit, Fox]

Math	Python	JavaScript
Set	{1,2,3}	new Set([1,2,3])
$\in$	x in A	A.has(x)
$\subseteq$	A <= B	manual function
$\cup$	`A`	B`
$\cap$	A & B	filter + has()

How does set theory apply to relational databases and search algorithms?

Can be used to filter data. For example mailing list subscribers that are subscribed to important notifications only.

Or patents of indian origin with a higher risk of diabetes.

Removes duplicates, e.g. users that downloaded the software more than once, or logged in from multiple machines.

Can be used for searching, e.g. articles about Trump in the last 24 hours.

Pages that have both keyword - are in both lists - intersections

Filtering, e.g. Clothing - shoes that are pink, in size 44 and trainers.

Appointments available today or tomorrow.

What role does set theory play in artificial intelligence and big data analytics?

Activity: Construct Venn diagrams to represent database queries and search engine indexing. Solve a real-world data structuring problem using set operations.

Advertise a garage to people in Grimsby:

Car Drivers $\cap$ People in Grimsby

Find all External Doors and Windows to mark fire escapes:

(External Doors $\subseteq$ Doors) $\cup$ Windows

Find the latest news about Trump

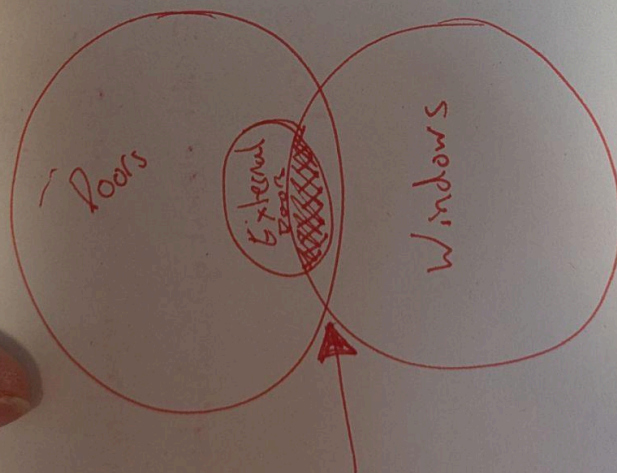
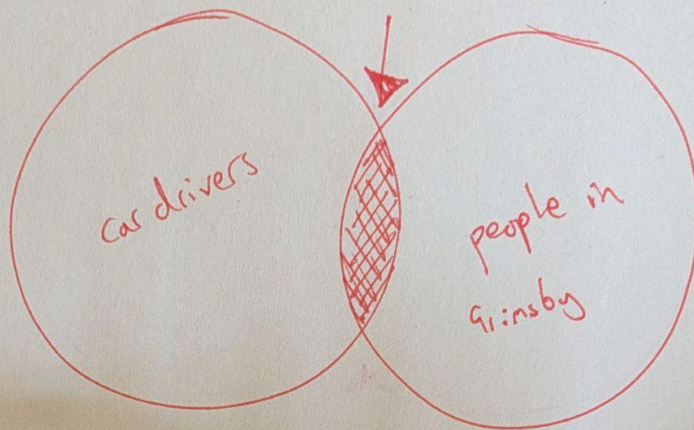
News articles about Trump $\cap$ From the last 24 hours

Drawings created in version 1 & 2:

Drawings created in version 1 $\cup$ Drawings created in version 2

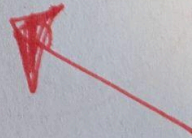
Patents that need a check up reminder

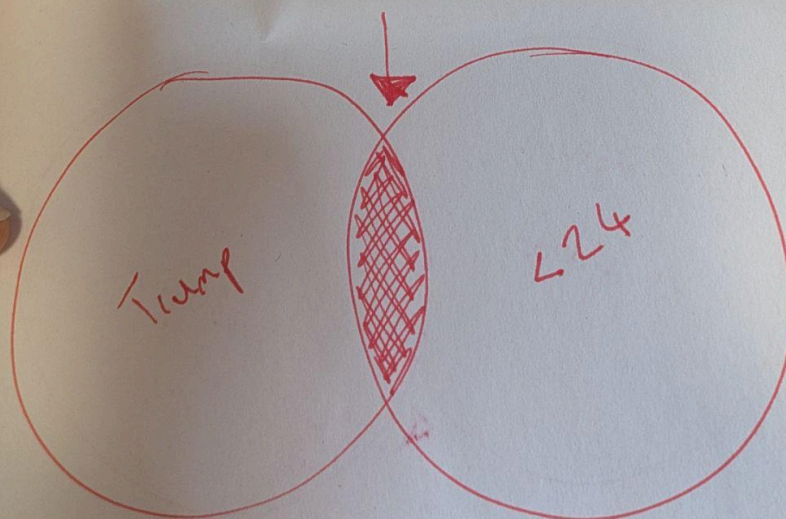
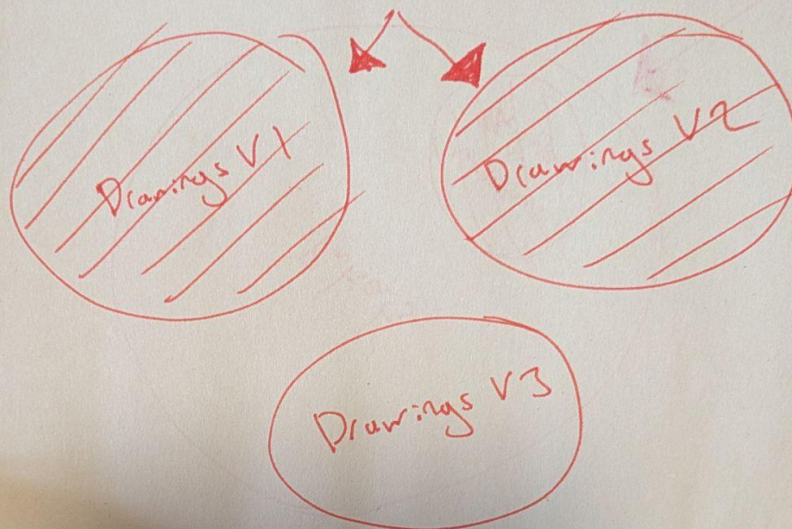
patents that have not had a check up in the last year. $\subseteq$ Patents



Patients

Subsidiary
or
clear





Notes

Good solution = time + accuracy + e

Vectors = ordered, same time, e.g. array

Lists = mixed items

Matrices 2D number/character arrays

Data frames = table of columns

References

OpenAI, 2026. Comparison of the Turing Colossus computer and a modern home laptop (HP Omen). ChatGPT conversation. Available at:
<https://chatgpt.com/share/698362bc-eeec-8008-8eb8-1a86938c6869>
(Accessed: 4 February 2026).

Randell, B. (1982). Colossus: Godfather of the Computer. In: Randell, B. (eds) The Origins of Digital Computers. Texts and Monographs in Computer Science. Springer, Berlin, Heidelberg. Available at: https://doi.org/10.1007/978-3-642-61812-3_27 (Accessed: 4th February 2026).